PPR Acquisition Policy Under Nonstationarity — October 16, 2026

Results generated 2026-09-30.

Project Overview

This project asks whether accounting for non-stationarity leads to better land-acquisition decisions for waterfowl conservation in the U.S. Prairie Pothole Region (PPR) under different climate scenarios. It builds three decision models and compares their performance across climate-change scenarios and a stationary null.

The three models acquire land period by period (decadal steps, 2020–2100) under a budget constraint:

- **Greedy** — status-quo heuristic: rank parcels by current benefit-to-cost ratio, buy down the list. No foresight.
- **Myopic ILP** — re-optimizes each decade but freezes current benefit and hazard and assumes they persist (a fabricated stationary future); implements the current decade, then re-solves.
- **Rolling-horizon ILP** — same structure, but uses the true projected trajectories. The foresighted benchmark. Uses Gurobi solver (academic license).

The spatial unit is the **EAU** (Equal Area Unit, ~282 km^2). After dropping 4 Wetland Management Districts, the analysis uses **879 EAUs across 20 WMDs**, over **9 decadal periods** and **5 scenarios** (RCP 4.5/8.5 × wet/dry, plus a stationary null).

Data Sources

- **Benefit**: McKenna, Owen P., et al. "Conservation enhances resilience of the United States Duck Factory as environmental shifts may diminish landscape-scale waterfowl breeding capacity." BioScience (2026): biag091.

  - linear interpolation to get from obs/mid-century/end of century to decadal time steps.
  - projected over RCP 4.5/8.5 × wet/dry scenarios.

- **Cost**: Nolte, Christoph. "High-resolution land value maps reveal underestimation of conservation costs in the United States.". Proceedings of the National Academy of Sciences. 117.47 (2020): 29577-29583.
  - assume 2% unfiform annual increase
  - note no emissions or precip scenarios.

- **Risk**: FOREsce (derived). Sohl, Terry L., et al. "Spatially explicit land-use and land-cover scenarios for the Great Plains of the United States." Agriculture, Ecosystems & Environment 153 (2012): 1-15.
  - calculated as net suitable habitat loss per time step.
  - note this has RCP 4.5 & 8.5 scenarios, but no wet/dry difference.

Model Notes

- **Acquisition value is calculated as expected loss prevented:**

  `V[i, ] = Σ_{t= }^{T}  ^t · b[i,t] · (1 - S[i,t])`.

  This is the value gained by removing conversion risk from parcel i starting at (i.e., value of acquiring a parcel). All unacquired parcels are treated as available at each re-solve; conversion risk does not remove parcels from the set, but instead enters **only** through the weighting in the objective. Acquisition removes this weight and counts full (discounted) value; unaquired parcels retain only their survivial-discounted value.

- **Budget**: set to a scalable amount based on a number of "median priced" EAUs per period. Currently set to the dollar equivalent of 5-median priced EAUs per period.

- **Discount Factor**: 0.95 decadal

Results

Descriptive Summary

Duck distribution

Consistent with other studies (e.g., McKenna et al, 2026), duck abundance is higher in the western portion of the PPR than the eastern portion. This overall distribution pattern continues through 2100 (Fig. 1). In all scenarios, duck abundance is predicted to decline across the PPR by 2100, although wet scenarios fare better than dry scenarios (Fig. 2).

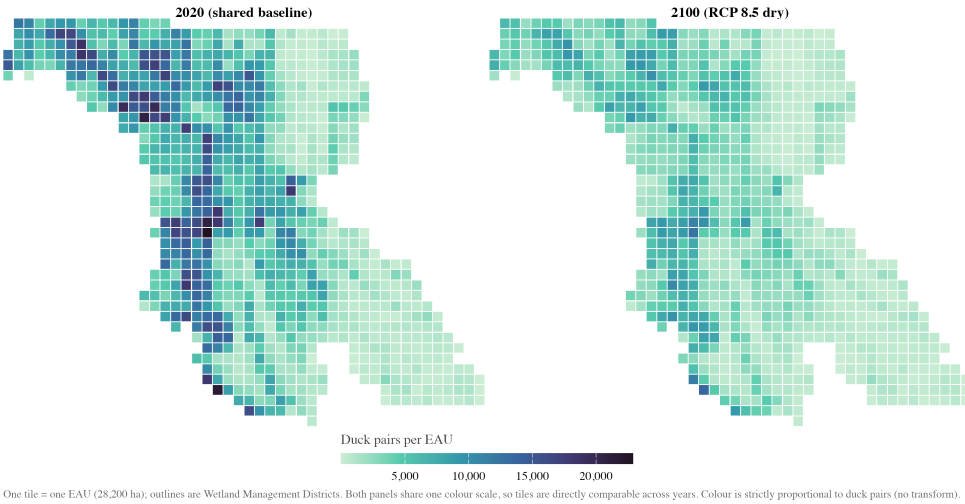


Figure 1: **Duck-pair abundance across the PPR: 2020 baseline vs. 2100 under RCP 8.5 dry.** The 2020 panel is the shared cross-scenario baseline common to all five scenarios — only the 2100 panel is scenario-specific. One tile = one EAU.

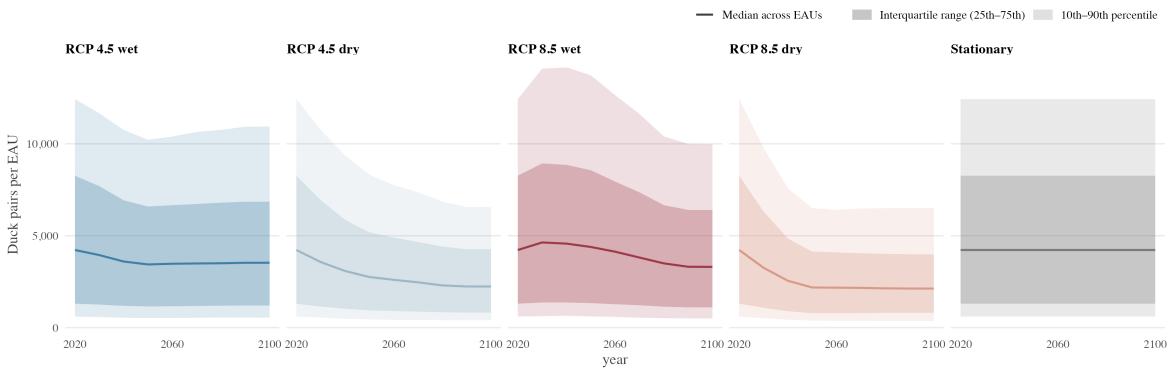


Figure 2: **Per-EAU duck-pair abundance to 2100, by scenario.** Solid line = median across EAUs; dark band = interquartile range; light band = 10th–90th percentile. Bands show spatial variability across EAUs within a given scenario-year.

Risk

Risk of conversion across all EAUs is heavily skewed, with the vast majority of EAUs showing a nearly zero per period conversion risk; the maximum per period conversion risk for a given EAU is ~15%. There is a sharp drop in per period conversion risk in 2050, which is a function of the FOREsce model: cropland conversion is frontloaded through 2050, at which point conversions to cropland decline.

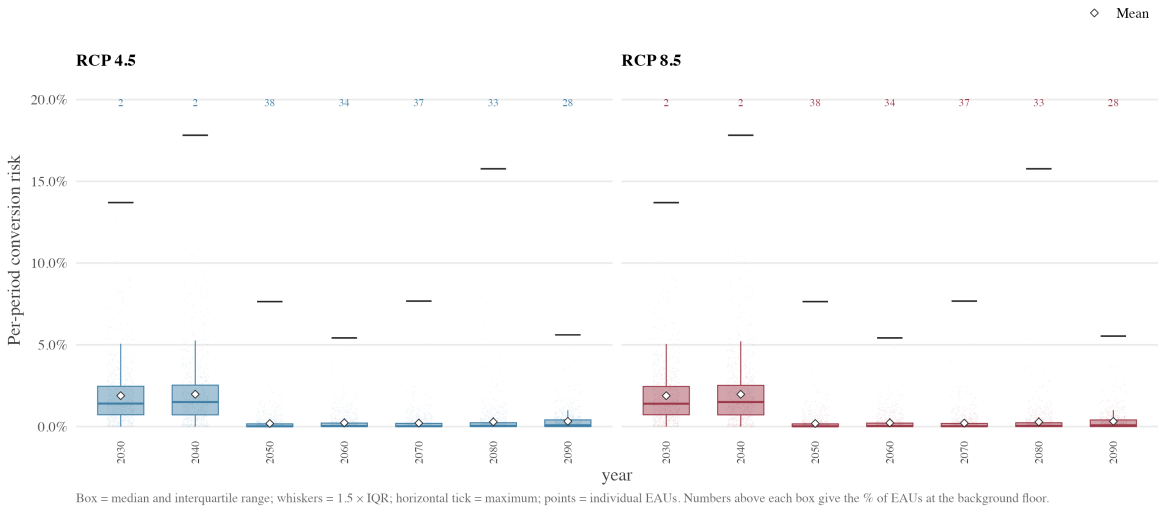


Figure 3: **Per-period conversion risk across EAUs, by decade and emissions scenario.** Box = median and IQR; whiskers = 1.5 × IQR; horizontal tick = maximum; points = individual EAUs. The number above each box is the % of EAUs sitting at the background conversion floor.

Cost

Acquisition cost per hectare in USD 2020. High cost parcels are concentrated in the eastern portion of the PPR, consistent with other analyses (e.g., Ando and Mallory, 2012).

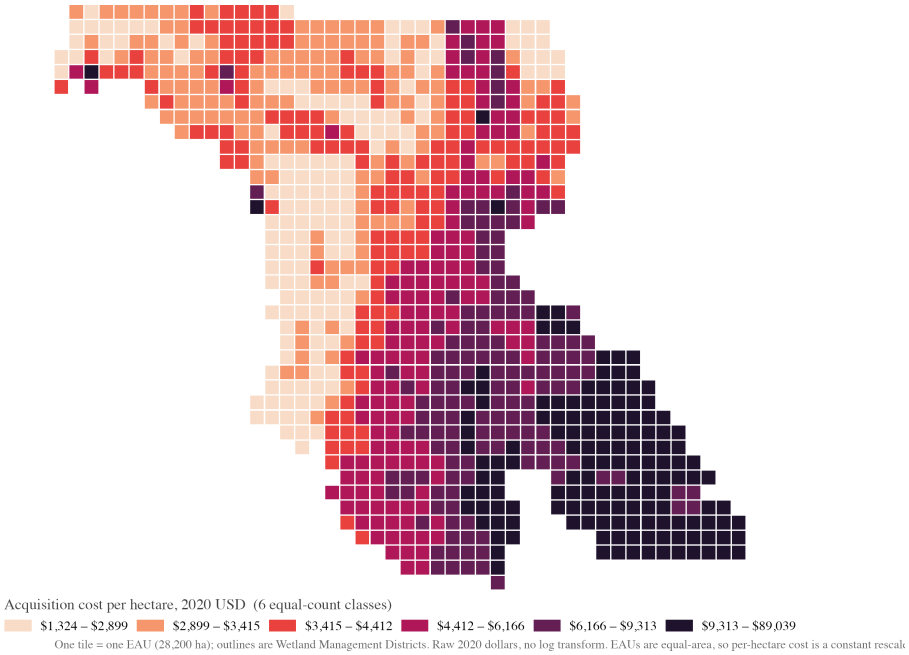


Figure 4: **Acquisition cost per hectare, 2020 USD.** Raw dollars, no log transform, grouped into six equal-count classes.

Model results

Acquisition value over baseline

This table compares the total benefit secured across the landscape (2020-2100) via three different acquisition policies (rolling, myopic, and greedy) against a "do-nothing" baseline scenario in which no parcels are acquired. All acquisition policies outperform the do-nothing baseline under climate change scenarios; there is no improvement under assumptions of stationarity (as expected: stationarity freezes conversion risk to effectively zero).

Even a greedy acquisition policy outperforms a do-nothing policy; myopic and rolling policies each respectively improve upon that baseline. As expected based on the duck density projections, wet/dry scenario is a larger determinant of improvement than RCP scenario (see Fig. 6).

Overall improvement of any acquisition policy, expressed in % over baseline, is relatively small (0.52% at best). This is likely an artifact of how conversion risk is being calculated in this model (a measure of habitat erosion standing in for habitat conversion probability). That said, these numbers might not be that different from real world estimates: Kemnick et al 2023 estimated that the program impact for SWAP was "an avoided loss between 0.00% and 0.02% of the carrying capacity for broods and breeding waterfowl from 2008–2017."

Table 1: **Value of acquisition policy over do-nothing baseline, by scenario.**

Scenario	Greedy		Myopic		Rolling	
	% over baseline	Δpairs	% over baseline	Δpairs	% over baseline	Δpairs
RCP 4.5 wet	0.40%	120,285	0.53%	156,775	0.64%	189,323
RCP 4.5 dry	0.36%	84,318	0.46%	108,656	0.54%	128,523
RCP 8.5 wet	0.40%	130,656	0.52%	171,625	0.62%	202,666
RCP 8.5 dry	0.35%	74,899	0.45%	94,835	0.53%	112,225
Stationary	<0.0001%	12	<0.0001%	12	<0.0001%	13

Gap: Foresight vs. Myopic/Greedy

This table compares the gap between total benefit secured across the landscape (2020-2100) in the greedy or myopic policy as compared to the rolling policy. The rolling policy is used as the provably near-optimal benchmark (MIP gap 0.01% across all rolling solves). As expected, the greedy policy displays a larger gap with the near-optimal rolling policy than the myopic policy under all climate scenarios. Myopic underperforms optimal by ~17-19%; greedy underperforms optimal at around ~30%. Under stationarity, there is virtually no difference (as expected).

This "value added gap" is the metric of greatest interest, since it describes the performance of near optimal conservation policy as compared to a greedy (i.e., status quo) or marginally improved policy. The rolling policy outperforms the myopic and greedy policies under all climate scenarios, consistent with our expectations that an acquisition policy with foresight should outperform a policy with no or limited foresight.

Highlight: A non-stationary, climate-aware acquisition policy would deliver a 33-36% improvement in conservation outcomes as compared to the greedy (i.e., status quo) acquisition policy in use today.

Table 2: **Value Added Gap: rolling vs. myopic/greedy, by scenario.**

Scenario	Greedy		Myopic	
	gap %	Δpairs	gap %	Δpairs
RCP 4.5 wet	36.47%	69,039	17.19%	32,549
RCP 4.5 dry	34.39%	44,205	15.46%	19,866
RCP 8.5 wet	35.53%	72,010	15.32%	31,041
RCP 8.5 dry	33.26%	37,326	15.50%	17,390
Stationary	2.92%	<1	1.30%	<1

gap % = shortfall as % of rolling's value-added.

Δpairs = rolling value-added − model value-added (discounted duck-pairs).

Value added gap over time

This graph shows how the gap between the rolling policy and the myopic/greedy policies grows over time. This is presented as net changes in duck pairs, using the rolling policy as a benchmark (value added gap for rolling = 0). As expected, the cumulative gap in performance between rolling and greedy/myopic grows over time (positive values only); and greedy is substantially worse than myopic (red always > yellow).

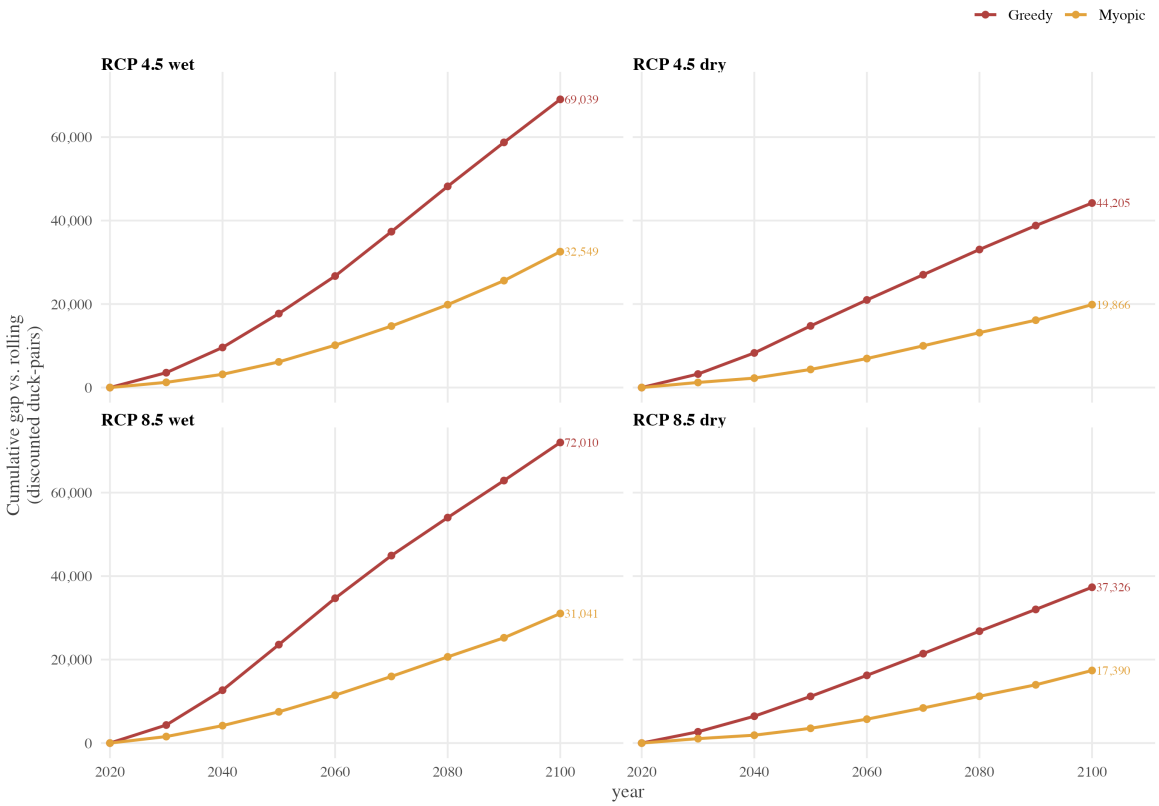


Figure 5: **Conservation shortfall: rolling − myopic/greedy.** Discounted duck-pairs (cumulative to 2100) not conserved as compared to rolling, by climate scenario. Rolling is the optimal benchmark (gap = 0); MIP gap ~0.01%.

Landscape decline

Here we show the rolling policy under stationarity and 4 climate scenarios, over time. Under all climate scenarios, landscape totals decline; dry scenarios fare worse than wet, whereas emissions scenario is less of a determining driver.

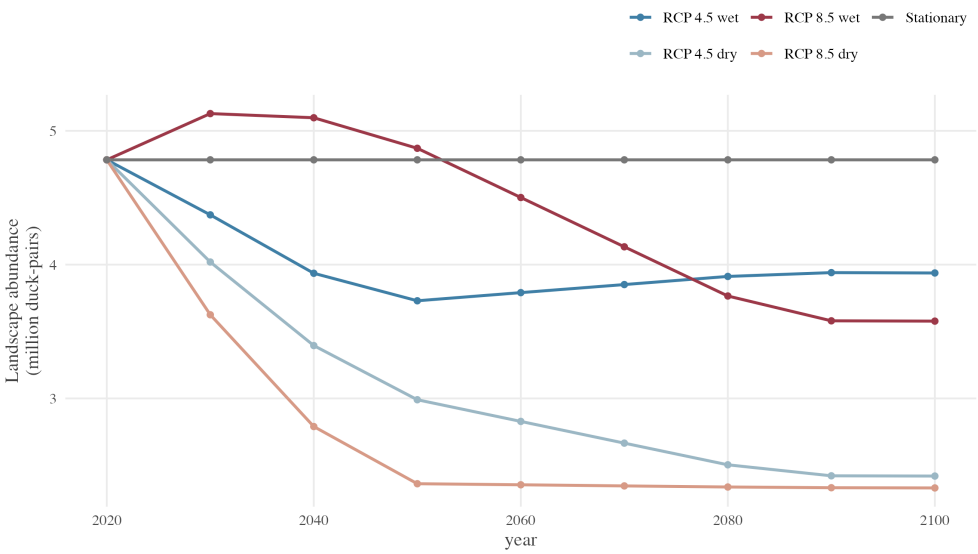


Figure 6: **Landscape decline under optimal acquisition schedule, by scenario.** All scenarios share a common 2020 anchor [4.76m]. Decline is sharpest in dry scenarios.

Acquisitions by policy: rolling vs myopic

This map shows parcels that are acquired under both rolling and myopic acquisition schedules; those that are acquired only under rolling; and those that are acquired only under myopic. Light blue parcels are those that are purchased only under an optimal acquisition policy; the myopic model skips these parcels. The orange parcels are ones that the myopic policy purchased but the rolling policy declined to purchase. Interpretation here is that light blue parcels are where the rolling policy's advantage comes from; orange is where the myopic dedicated funds sub-optimally.

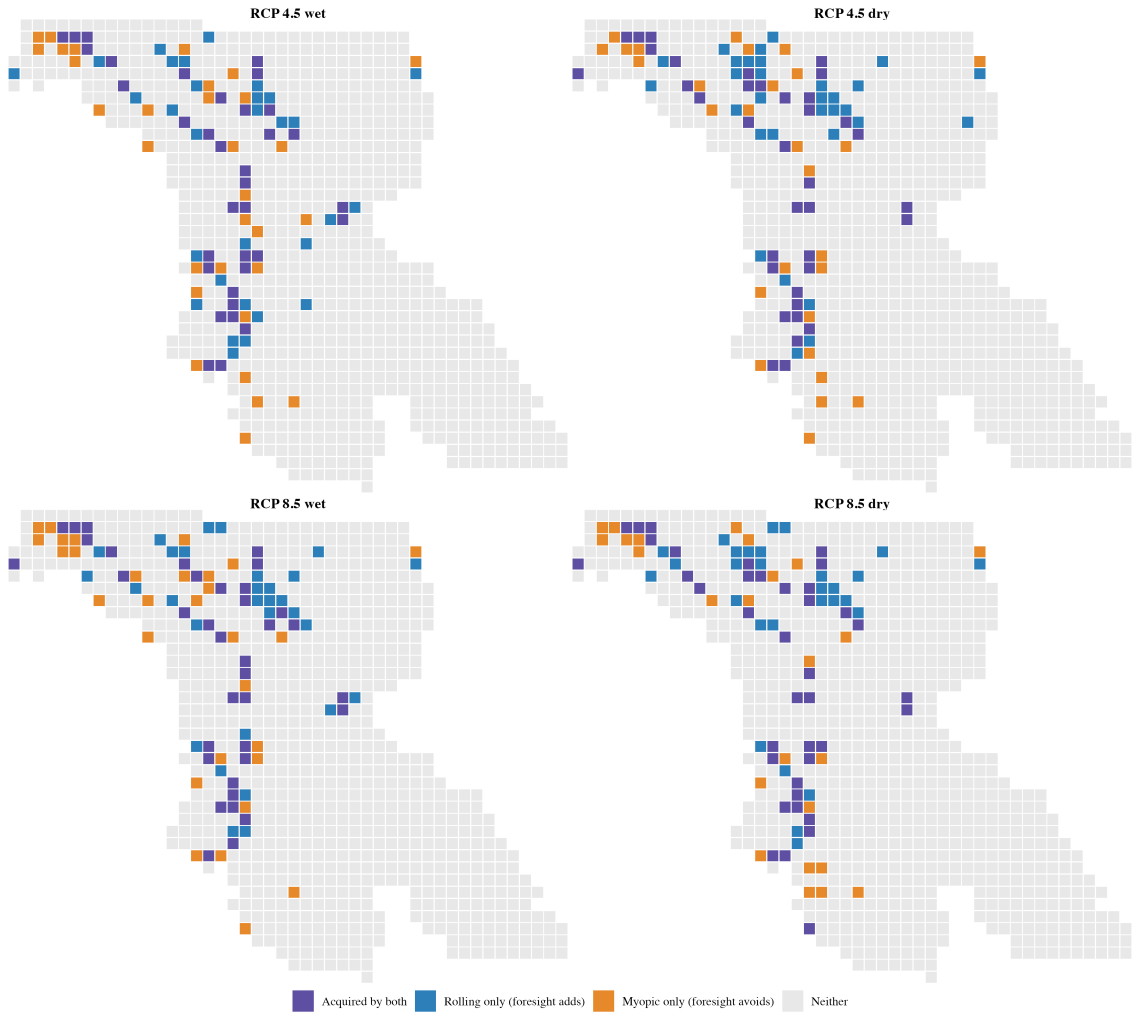


Figure 7: **Rolling vs. myopic acquisition sets.** EAUs colored by which policy buys them, under each climate scenario. Purple cells are parcels acquired under both policies; blue cells are purchased only under the rolling policies; orange cells are purchased by the myopic policy but avoided by the rolling policy.

Value added gap by policy: rolling vs myopic

This map shows how rolling and myopic acquisition schedules differ in where and how they prevent duck loss. Cells are highlighted where myopic and rolling schedules differ in acquisition decisions. Blue cells show where the rolling acquisition schedule acquired parcels but myopic did not; this represents addition ducks secured through foresight. Red cells show where the myopic schedule acquired parcels that the rolling schedule ignored; this represents a sub-optimal allocation of budget. Darker shades of blue are parcels with higher duck density. Darker shades of red indicate parcels where the difference between realized and potential loss is greater.

Take away: this map show the spatial distribution of how the two models allocated their identical budget differently. The rolling model invested in parcels that maximized benefit by 2100; the myopic invested in parcels that secured real duck pairs (no loss, only gain), but underperformed the rolling benchmark.

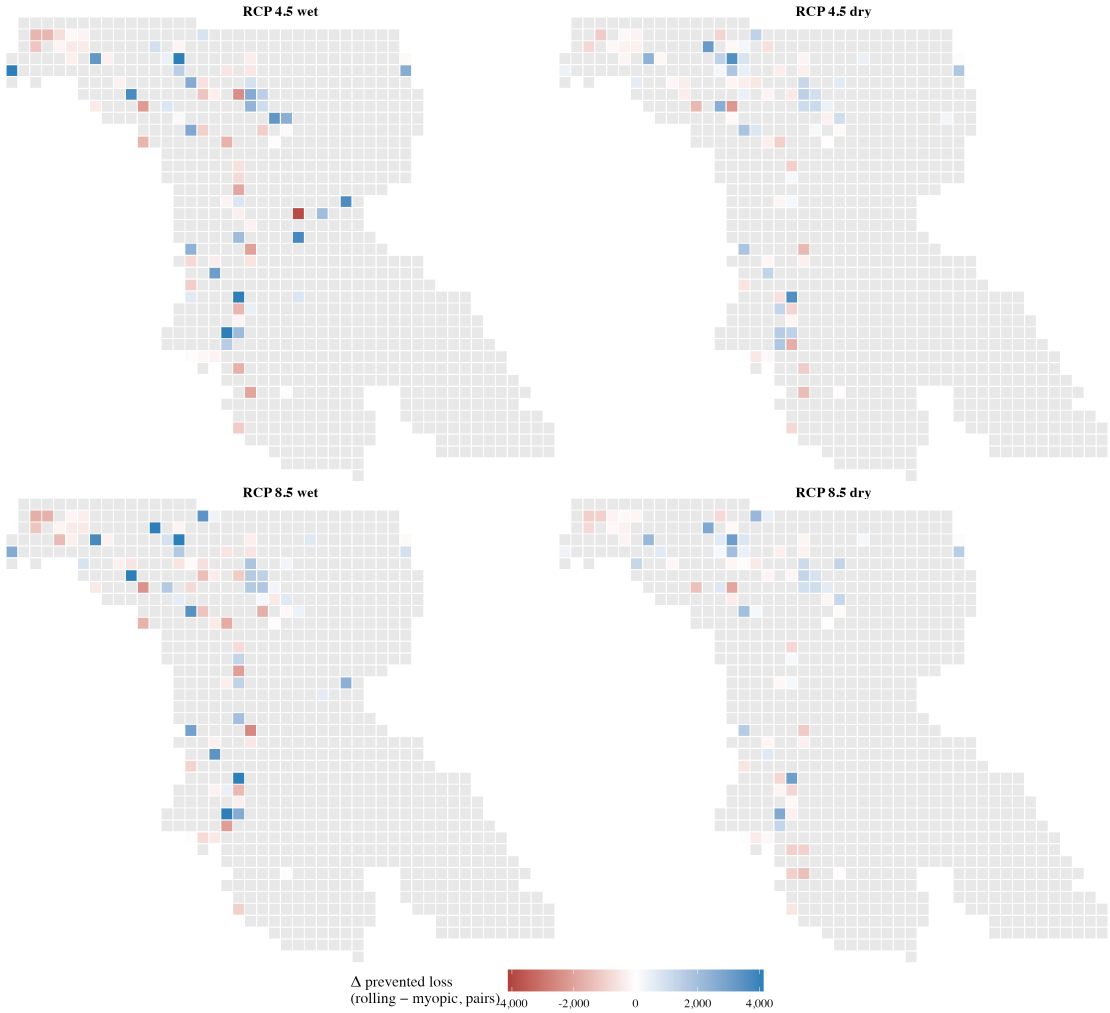


Figure 8: **Per-parcel prevented-loss: rolling — myopic.** Blue: rolling invests here and prevents duck-pair loss; red: myopic invests here instead, securing absolute conservation benefit but relatively less than rolling. grey: no difference Coloured cells sum to the value-added gap.

Updates

9/30/2026: Protected Area Removal

In examining the conversion risk, we realized that the previous conversion risk included protected area in the total pool of potentially available habitat, which artificially diluted the estimated conversion risk. To address this, I overlayed the USGS Protected Area Database (PADUS) and calculated the gross suitable habitat loss with protected areas removed. This revised gross suitable habitat rate was used in the risk of loss calculation (i.e., script 05); however, the other uses of landcover data (such as distributing waterfowl density within a WMD) continue to use the unmasked raster with protected areas included.

To overlay the PADUS layer, I used the PAD-US 4.1 Vector Analysis (USGS GAP, 2024, PAD-US 4.1, https://doi.org/10.5066/P96WBCHS) and masked out polygons marked as GAP Status Code 1-3 (which covers areas with permanent protection, like National Parks, and areas where "low intensity" land uses is allowed). I then created a 0/1 raster of protected land within the PPR, and removed those areas when calculating the gross habitat loss.

9/15/2026: Conversion Risk Update: Net vs Gross Habitat loss:

Delving in to my calculations of how I quantify conversion risk, I realized that I was using net rather than gross habitat loss- this was not conceptually consistent with what I presented, which I intended to be the proportion of suitable habitat in time t that was lost by time t+1 (a flow). Instead, I had calculated the difference between the stock of suitable habitat in time t - t+1, which could be altered by any gains in suitable habitat.

I changed the calculations to see how this would affect the estimate of conversion risk, and it was fairly substantial- there is a fair amount of habitat recovery in the FOREsce models, so some of those habitat gains were concealing real habitat losses. The table below compares the net vs gross habitat loss rate (gross > net in all decades but particularly mid-century). In addition, the use of the gross metric in the first model run meant that some EAUs had a "positive" habitat loss (i.e., overall habitat gain), which I corrected for by imposing a data-derived epsilon (floor). With the net metric, that is no longer needed (net loss >= 0). The table also shows the percent of EAUs that were "censored" by the artifical floor (i.e., which really had habitat loss but which was hidden by habitat gain).

Decade	gross	net	% of EAUs censored
2020–2040	~0.018	~0.017	0.5–7%
2050–2070 (4.5)	~0.002	0.0009	**14–17%**
2050–2070 (8.5)	~0.002	−0.0009	**16–24%**
2080–2090	~0.003	~0.002	~4%

7/30/2026: Spatial Join error

I discovered an error in the spatial join of the extracted land cover data and the model's data panel. I realized that the landcover data was joined to EAUs by row position, which introduced errors- land classification (% suitable habitat) was joined to the wrong EAUs. Conversion risk (hazard) was the variable most affected- the trans_prob were not attached to the correct spatial unit. Because duck density is distributed according to % suitable, it also meant that duck distributions were incorrect- but the scale of this error was mitigated by the fact that WMD-scale determines the absolute amount of duck density. This meant that WMD level abundance numbers are fine, but distribution within WMD to EAUs was incorrect; cost numbers are unaffected.

Because of this, I had incorrectly believed that most of the EAUs with missing landcover data were concentrated in Winsom, which is why I removed it- but I discovered that this wasn't actually the case. There were in fact multiple WMDs with at least a few EAUs missing data. I decided to make a cap of 20% (any WMD missing 20% of EAU land cover data would be excluded); the others have slightly inflated duck numbers in their remaining EAUs.

A few interesting findings: - spatial distribution of duck abundance is now more clearly aligned with McKenna 2026 (good news!). Compare Fig. 1 with Fig. 1 from "results_07302026" - there is now a clearer alignment of duck abundance and LOW acquisition cost: this is more consistent with the literature, and also means that greedy starts performing slightly better than before. (gap in value add vs rolling =20%; it was about 40% before) - but interestingly, the value added of rolling vs myopic increases: we are now looking at ~20% improvement, not ~15%.

7/10/2026: Temporal alignment of benefit data

After the initial model run, I realized there was a problem with the temporal alignment of the benefit data with the landcover data (and the decision model itself). The benefit data is as follows:

- "Obs" = mean 2007–2017
- "3059" = mean 2030–2059
- "7099" = mean 2070–2099

Originally, I had aligned it as follows:

- "Obs" = mean 2007–2017 -> 2020
- "3059" = mean 2030–2059 -> 2050
- "7099" = mean 2070–2099 -> 2100

After talking with Owen, I updated the alignment to use window midpoints :

- "Obs" = mean 2007–2017 -> 2012
- "3059" = mean 2030–2059 -> 2045
- "7099" = mean 2070–2099 -> 285

Notes: The 2020 baseline is produced by interpolating each RCP×GCM combo on its 2012→2045 segment and then averaging across the four combos into one shared period-0 row (preserves the shared-baseline used for the 2020 risk). Decades 2030/2040 interpolate on the 2012→2045 segment; 2050–2080 on the 2045→2085 segment; 2090 and 2100 are held constant at the 2085 value

I re-ran the model with this midpoint temporal alignment of benefit data; saw little difference in model results. My takeaway is that we can reasonably use this midpoint alignment, and importantly that it doesn't much matter for model results.

Table 3: **Temporal alignment of benefit data: midpoint comparison.**

Scenario	Greedy		Myopic		Rolling	
	Δ % over baseline	Δpairs	Δ % over baseline	Δpairs	Δ % over baseline	Δ Δpairs
RCP 4.5 wet	+0.02%	+2,600	+0.01%	+305	+0.02%	+885
RCP 4.5 dry	0.00%	-3,742	+0.01%	-5,335	+0.01%	-8,471
RCP 8.5 wet	-0.01%	-6,610	0.00%	-8,285	-0.01%	-11,626
RCP 8.5 dry	+0.01%	-1,135	+0.02%	-2,103	+0.02%	-2,965
Stationary	0	-1	0	-1	0	-1

Δ = change vs. the primary benefit alignment (midpoint comparison).

Table 4: **Temporal alignment of benefit data: gap vs. rolling (midpoint comparison).**

Scenario	Greedy		Myopic	
	Δ gap %	Δ Δpairs	Δ gap %	Δ Δpairs
RCP 4.5 wet	-1.00%	-1,714	+0.22%	+581
RCP 4.5 dry	-0.62%	-4,731	-1.38%	-3,136
RCP 8.5 wet	+0.13%	-5,015	-0.74%	-3,341
RCP 8.5 dry	-0.44%	-1,829	-0.34%	-861
Stationary	-0.56%	0	+0.21%	0

Δ = change vs. the primary benefit alignment (midpoint comparison).